

The Capstone: Supermarket Sales Operations Analysis

Scenario: You are the lead Data Analyst for a global supermarket chain. Management has provided a messy data extract of the last 1,000 transactions across three branches. You have 3 hours to clean the database, establish baseline financial metrics, build a visual dashboard, and use probability distributions to model future store operations. Most importantly, management expects you to interpret your findings.

Part 1: Data Cleaning (Power Query)

Import the raw CSV into Power Query and perform the following transformations. Load the result into a new sheet called "Clean_Data".

1. **Split Columns:** Split City_CustType into two separate columns using the correct delimiter. Name them City and Customer Type.
2. **Standardize Text:** The Product line column has inconsistent capitalization. Force all text in this column to "Capitalize Each Word".
3. **Handle Nulls:** Filter out any transactions where the customer Rating was left blank.
4. **Data Types:** Ensure Unit price and Total are formatted as Currency, and Quantity is formatted as a Whole Number.

Part 2: Baseline Metrics & Margin of Error

Using your "Clean_Data" sheet, calculate the following on a new "Stats Summary" sheet:

1. **Central Tendency:** Calculate the **Mean**, **Median**, and **Mode** for the Total transaction value.
2. **Dispersion:** Calculate the **Variance.S** and **Standard Deviation.S** for the customer Rating column.
3. **Margin of Error:** Take a random sample of 60 transactions. Calculate the **Standard Error** and **Margin of Error** for the Total sales value, assuming a 95% confidence level (Z-score = 1.96).
4. **Observation 1:** Look closely at your Mean and Median for the Total sales. Are they perfectly equal, or is one noticeably higher than the other? Based on your knowledge of distributions, what does this tell you about the shape of the store's sales data (is it perfectly symmetrical, or skewed by a few massive purchases)?

Part 3: Visualizing the Operation (The Dashboard)

Create a new worksheet named "Visuals". Build the following charts, ensuring each has a proper title and data labels.

1. **XY Scatter Graph:** Plot Quantity (X-axis) against the Total bill (Y-axis). Add a linear trendline.
2. **Column Bar Chart:** Create a bar chart showing the sum of Total revenue generated by each of the three Branch locations (A, B, and C).

3. **Pie Chart:** Create a pie chart showing the proportion of transactions broken down by Payment method (Cash, Credit Card, Ewallet). Format the data labels to show Percentages.
4. **Histogram:** Create a Histogram of the customer Rating column to visualize how satisfaction scores are distributed.
5. **Observation 2:** Look at your Bar Chart and your Pie Chart. If the marketing team has a limited budget to run a promotion for "Ewallet" users, which Branch should they target to get the highest overall revenue impact, and why?

Part 4: Operational Modeling (Distributions)

Use the exact descriptive statistics calculated in Part 2, alongside the raw data, to solve these probability scenarios using Excel formulas.

The Normal Distribution (Sales Revenue)

- **Task A:** Using your specific Mean and Standard Deviation for the Total column, what is the probability that the next customer's total bill will be **\$500 or less**?
- **Task B:** What is the probability that a customer's bill will be **greater than \$800**?
- **Observation 3:** Based on these two probabilities, if the store manager has to choose between stocking up on expensive high-ticket items or focusing on volume for smaller baskets, what does the math suggest they do?

The Binomial Distribution (Payment Methods)

- **Task A:** Look at your Pie Chart to find the overall percentage (probability of success) for "Credit card" transactions. If 50 customers walk into the store in the next hour, what is the exact probability that **exactly 20** of them will pay with a Credit card?
- **Task B:** What is the probability that **exactly 0** out of the next 10 customers use a Credit card?

The Poisson Distribution (Foot Traffic)

- **Task A:** Determine the average number of transactions per day (Lambda) by counting your total rows and dividing by the number of unique dates in the dataset. Using this average, what is the probability that the store will process **exactly 85 transactions** tomorrow?
- **Task B:** What is the probability of processing **exactly 120 transactions** tomorrow?
- **Observation 4:** Compare the results of Task A and Task B. If the probability of getting 120 transactions is extremely low, how should the store manager use this information when creating the employee shift schedule?

The Exponential Distribution (Equipment Failure)

- **Task A:** The IT department notes that the receipt printers jam on average every 45 minutes (Lambda = 45). What is the probability that the next jam occurs in **20 minutes or less**?

- **Task B:** What is the probability that the printer runs smoothly for **more than 60 minutes** before jamming? (*Hint: Remember the rule of 1 minus the cumulative probability*).

The Uniform Distribution (Promotions)

- **Task A:** Management is giving a \$50 voucher to one random receipt number today. If there are exactly 300 receipts generated today, what is the uniform probability of receipt #142 winning the voucher?
- **Task B:** Does receipt #299 have a higher, lower, or equal probability of winning compared to receipt #142?

Part 5: Hypothesis Formulation

You do not need to solve these mathematically, but management needs you to structure the business tests.

1. **Test 1:** The marketing team believes that customers who hold a "Member" card spend more money on average than "Normal" customers.
 - State the **Null Hypothesis**.
 - State the **Alternative Hypothesis**.
 - Identify if this is a **directional (one-tailed)** or **non-directional (two-tailed)** hypothesis, and briefly explain why.
2. **Test 2:** The regional manager wants to know if there is any difference whatsoever in the average customer Rating between Male and Female shoppers.
 - State the **Null Hypothesis**.
 - Identify if the Alternative Hypothesis would be directional or non-directional.